Timing-aware clustering

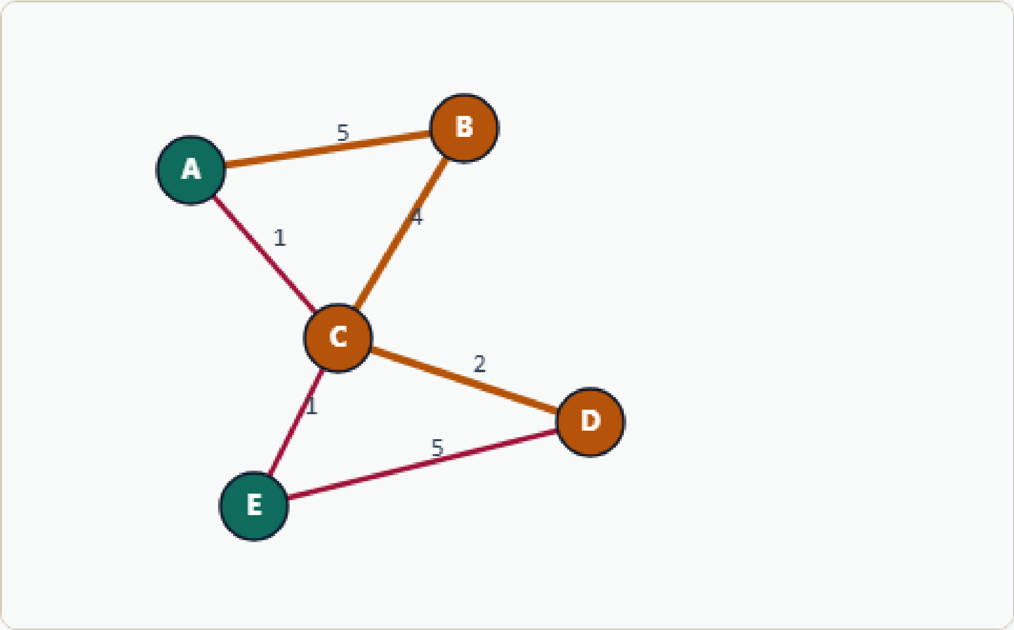
Critical nets should stay uncut

Mark critical edges

TIMING-AWARE CLUSTERING

Mark critical edges

Step 1 / 5 · criticality · module04-03-timing-aware-clustering



Explanation

Criticality map emphasizes path A–B–C–D: A–B=5, B–C=4, C–D=3. Cutting critical edges hurts timing more than raw wire weight suggests.

- weighted edge = w × criticality
- Start from BAD_SEED
- FM on the weighted graph

```
A|B=5
B|C=4
C|D=3
```

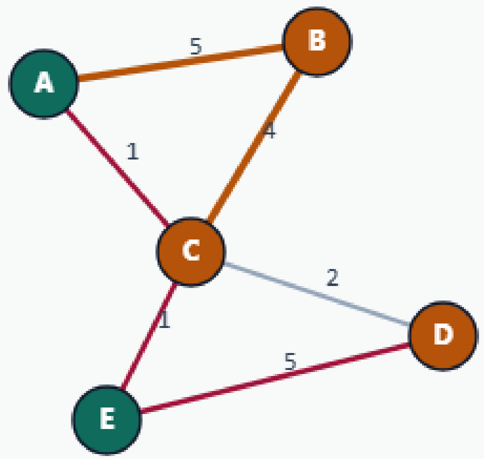
Mark critical edges

Reweight then refine

TIMING-AWARE CLUSTERING

Reweight then refine

Step 2 / 5 · weight · module04-03-timing-aware-clustering



Explanation

Multiply each edge by its criticality (default 1 if missing). FM now strongly prefers keeping A–B and B–C internal.

- Protect the critical path
- Still a bipartition FM
- Report plain and weighted cuts

```
engine: FM on weighted edges
```

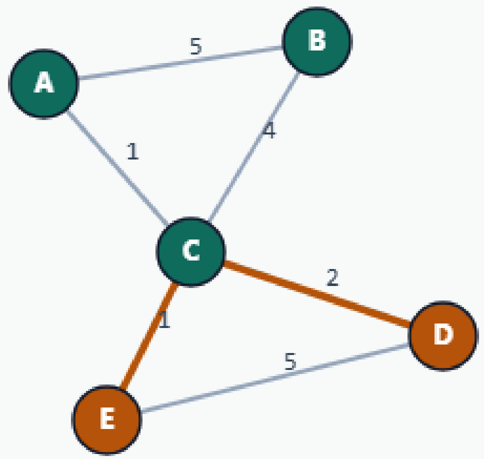
Reweight then refine

Land on ABC|DE

TIMING-AWARE CLUSTERING

Land on ABC|DE

Step 3 / 5 · result · module04-03-timing-aware-clustering



Explanation

Timing-aware FM reaches ABC|DE. Plain cutsize is 3; weighted cut (criticality-scaled) is 7.

- plain=3
- weightedCut=7
- parts: ABC|DE

```
plain: 3
weighted: 7
```

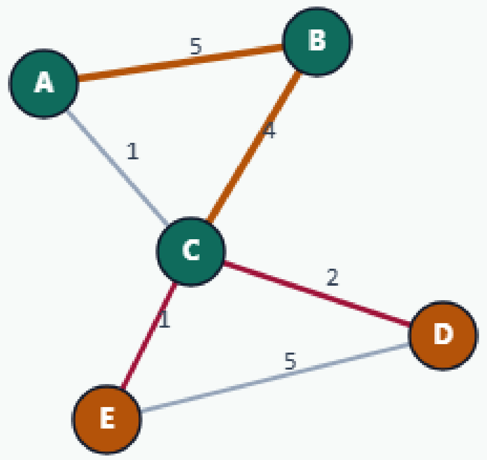
Land on ABC|DE

Critical edges uncut

TIMING-AWARE CLUSTERING

Critical edges uncut

Step 4 / 5 · protect · module04-03-timing-aware-clustering



Explanation

A–B and B–C stay inside ABC. The cut uses less critical bridges C–D and C–E — acceptable plain cut, better timing story.

- A–B (crit 5) internal
- B–C (crit 4) internal
- Bridge cut carries lower timing risk

```
critical path protected
```

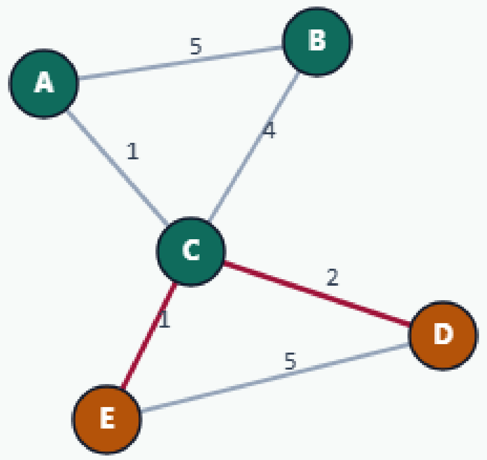
Critical edges uncut

Timing as an objective

TIMING-AWARE CLUSTERING

Timing as an objective

Step 5 / 5 · takeaway · module04-03-timing-aware-clustering



Explanation

Reweighting turns the same FM kernel into a timing-aware partitioner. Students compare plain vs weighted metrics on one seed.

- Criticality ≠ congestion
- Same BAD_SEED, different map
- Goldens: plain 3 / weighted 7

```
Starter golden: ABC|DE
```

Timing as an objective

Browser lab track

- In the browser lab, run timing-aware refinement from the bad seed
- Confirm plain cut three and weighted cut seven

Implement track

- Run timing-aware mode and confirm ABC versus DE, plain three, weighted seven

Implement track — try these

```
export PYTHONPATH=../common

python ../common/solvers.py examples/tiny_graph.json --mode timing
--seed ../module02-05-kernighan-lin/examples/seed_partition.json
--criticality examples/criticality.json
```

Pitfalls to watch

- Criticality of one means no boost, document defaults
- Mixing plain and weighted metrics in the same table confuses compares

Your turn

- Match the golden